

Implementing Mechanistic Interpretability Methods for Multilingual LLM Analysis

Unit: FOSE7901 – STEM Research Placement

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Project Pitch

Large language models (LLMs) can produce strong results while the internal processes leading to those results remain difficult to understand. This is especially important in multilingual settings, where equivalent prompts may produce different answers, confidence levels, or internal representations across languages. Looking only at the final response can identify these differences, but it does not explain where they arise inside the model.

Mechanistic interpretability provides methods for examining these internal processes. Sparse autoencoders identify interpretable features within model activations (Huben et al., 2024). The Jacobian Lens identifies representations associated with concepts that the model can potentially verbalise or use during internal reasoning (Gurnee et al., 2026). Activation patching provides a causal test by replacing internal activations and observing whether the model's behaviour changes (Dumas et al., 2025).

This placement will form **Phase 1** of a broader research plan. The focus will be to re-implement selected experiments from these three approaches, understand how each method works in practice, and test them on a small common set of prompts using one open-source LLM. The aim is to determine what each method reveals about internal model behaviour and whether the methods can later be combined into a practical mechanistic interpretability workflow.

This placement builds on my previous literature review and early work with sparse autoencoders and JLens. It will extend that foundation through structured implementation, causal intervention, and comparison of the three methods. The experience will directly support later research into multilingual and cross-cultural LLM behaviour.

The project relates to **UN Sustainable Development Goal 10: Reduced Inequalities**, as understanding language-dependent differences in LLM behaviour may contribute to more consistent AI systems across linguistic and cultural groups. It also relates to **SDG 9: Industry, Innovation and Infrastructure** through the study of methods that make AI systems more transparent and understandable.

1 Proposed Activities

The placement will be a computational research project using university or research-group computing resources, Google Colab, and hosted services such as Hugging Face where appropriate. Experiments that require access to internal activations will be run in environments where the model can be loaded directly.

The main research question is:

How do sparse autoencoders, the Jacobian Lens, and activation patching differ in the internal representations and causal mechanisms they reveal from the same controlled LLM prompts?

A secondary goal is to identify which method, or combination of methods, is most useful for later multilingual and culturally focused LLM research.

1.1 Baseline Method Review

I will first revisit the three baseline papers with a focus on the experiments selected for implementation. The purpose is not to conduct another broad literature review, but to understand the practical requirements of each method before applying it.

For each paper, I will identify the model component being analysed, the form of internal representation used by the method, the main output or intervention, the required software, and the main limitations that may affect implementation on a different model.

1.2 Sparse Autoencoder Implementation

The first method will follow Huben et al. (2024). Sparse autoencoders are designed to decompose dense model activations into a larger set of sparsely active features. The motivation is that an individual neuron may respond to several unrelated patterns, while a sparse feature may correspond more closely to a single interpretable concept or behaviour.

I will implement a selected sparse autoencoder experiment using the published code or a compatible implementation. Internal activations will first be collected from the chosen transformer model at selected layers. These activations will then be passed through a sparse autoencoder so that the original activation can be represented using a smaller number of active features.

The analysis will focus on which features become strongly active for selected prompts and what kinds of input patterns are associated with those features. I will examine whether features that respond to similar prompts also appear semantically related and whether particular features consistently appear for a specific type of model behaviour.

Where technically feasible, I will also modify or remove selected feature activations and measure whether this affects the model’s output. This will provide an initial indication of whether the identified features are only descriptive or also have a causal role in the behaviour being studied.

1.3 Jacobian Lens Implementation

The second method will follow Gurnee et al. (2026). The Jacobian Lens, or JLens, is designed to identify representations in intermediate layers that are related to concepts the model is positioned to verbalise later.

Unlike a standard logit lens, which directly applies the final unembedding operation to intermediate activations, JLens estimates how changes to an intermediate activation would propagate through the remaining layers. This allows concept-level information to be examined earlier in the model’s processing.

I will apply a fitted Jacobian Lens to selected prompts and record the highest-ranked concept or token readouts across model layers. This will allow me to observe whether a concept appears only near the final output or becomes visible earlier as part of the model’s internal computation.

The main analysis will examine how the JLens readout changes through the model, whether intermediate concepts appear even when they are not present in the final answer, and whether similar prompts produce similar internal concept trajectories.

I will also test simple JLens interventions such as strengthening, suppressing, or swapping selected concept directions. The resulting change in model output will help assess whether concepts identified by the lens are actively involved in the computation rather than simply correlated with it.

1.4 Activation Patching Implementation

The third method will follow Dumas et al. (2025). Activation patching tests causal relationships by taking an internal activation from one forward pass and inserting it into another.

I will construct paired source and target prompts and record the source model activations at selected layers. These activations will then be inserted into the target forward pass one layer at a time. Changes in output probability, predicted answer, or confidence will be measured after each intervention.

This will provide a direct way to identify which layers carry information that is important for the final model behaviour and will also provide a causal comparison with the representations identified by the sparse autoencoder and JLens approaches.

1.5 Small Common Example

A simple multilingual example will be used to show how the three methods can examine the same model behaviour. Figure 1 shows an illustrative case where equivalent English and Persian prompts about the cultural meaning of a moustache may activate different internal concepts.

For example, concepts such as *tradition* may appear in both languages, while another concept such as *war* may be represented more strongly in one language. Sparse autoencoders can be used to identify relevant internal features, JLens can show how these concepts appear across model layers, and activation patching can test whether changing the associated internal representation affects the model’s response.

The figure is an illustrative example only; the purpose of Placement 1 is to implement and compare the methods before applying them to larger datasets in later work.

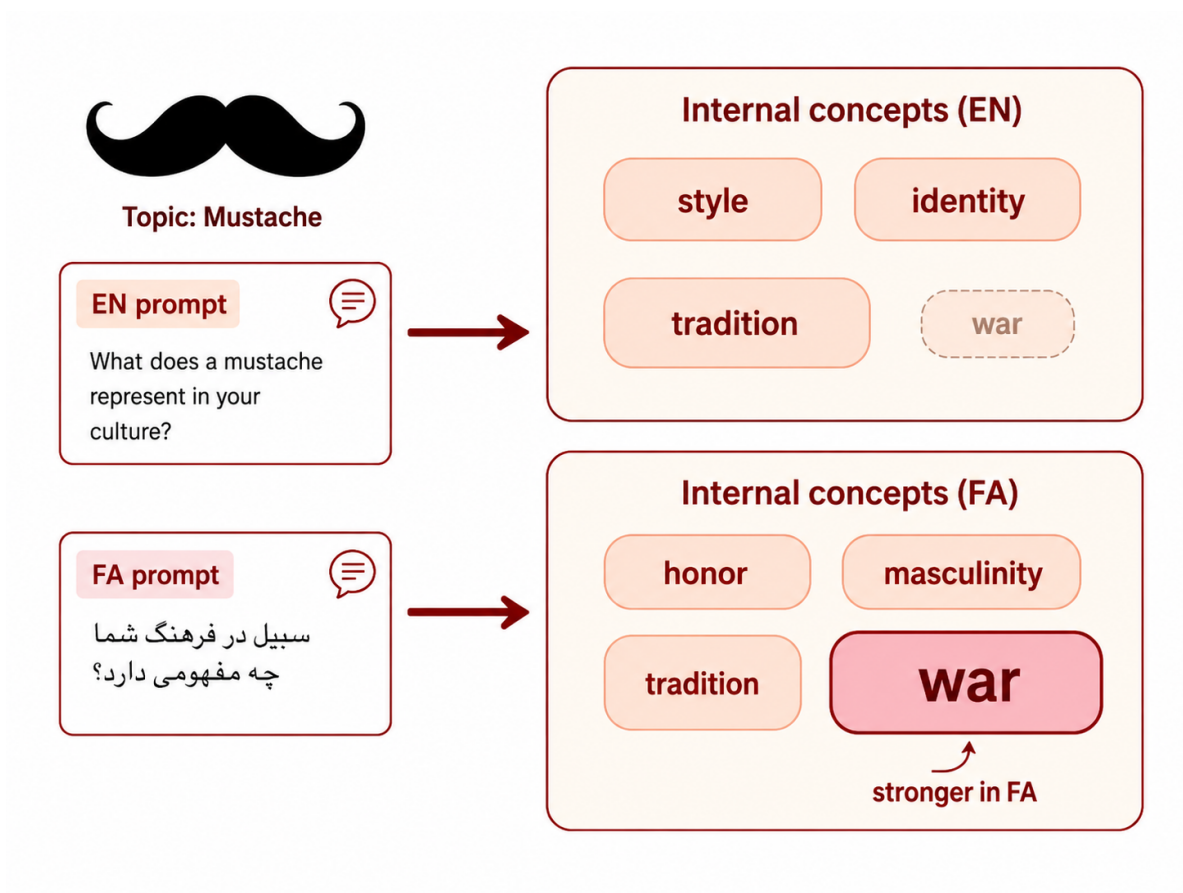


Figure 1: Illustrative example of how equivalent prompts in two languages may be associated with different internal concepts. The displayed concepts are hypothetical and are used only to demonstrate the planned analysis.

1.6 Controlled Prompt Set

After the baseline implementations are working, I will construct a small prompt set for comparison. The prompts will include semantically matched questions, simple multilingual variants, cases requiring an intermediate inference, and controlled prompt pairs suitable for causal intervention.

LLMs may be used to help generate candidate prompts, but the final prompts will be manually reviewed and edited before use.

The prompt set will remain deliberately small, approximately 30–100 prompt pairs depending on computational requirements. The emphasis is on detailed internal analysis rather than dataset size.

1.7 Common Model

The main analysis will use one open-source model in approximately the 7–9 billion parameter range. The preferred model is **Gemma 2 9B**, subject to computational availability and compatibility with the three methods.

If one of the implementations is not compatible with the preferred model, a similar open model supported by the original code may be used for that implementation. Any such change will be discussed with my mentor and recorded in the research log.

Using one main model will keep the project feasible and make comparison between methods more consistent.

1.8 Method Comparison

The three methods will be compared based on the type of internal representation they reveal, the layers they identify as important, how interpretable their outputs are, whether they provide descriptive or causal evidence, their computational requirements, and their suitability for later multilingual analysis.

The aim is not necessarily to choose one method as the best. The placement will instead determine how the methods complement one another and whether they can be combined into a stronger workflow for later research.

1.9 Supervision and Resources

Dr. Usman Naseem will provide guidance on project scope, experimental design, feasibility, and interpretation. Technical advice may also be sought from other researchers where required.

The main resources are summarised below.

1.10 Future Placement Direction

This FOSE7901 placement will focus on Phase 1: implementing and comparing the three baseline mechanistic interpretability methods on a small controlled prompt set.

Resource	Use
Python and PyTorch	Model execution, intervention, and analysis
Hugging Face Transformers	Loading open-source language models
Published SAE code	Sparse feature analysis
JLens implementation	Jacobian Lens analysis and intervention
NNsight	Activation access and patching where required
Google Colab / GPU resources	Running models and computational experiments
Git / GitHub	Version control and research records
University-approved storage	Saving code, results, and project records

A second placement could form **Phase 2**. This would focus on identifying suitable public datasets relevant to multilingual, cultural, moral, or regional LLM behaviour. The datasets would be collected, reviewed, cleaned, modified where required, and organised into consistent categories that can be used for mechanistic interpretability experiments.

The combined interpretability workflow developed during Phase 1 could then be applied to these larger and more structured datasets. This would allow the later project to move from learning and validating the methods to using them for a broader research question.

2 Induction / Training Requirements

This project is entirely computational and does not involve laboratory work, fieldwork, hazardous materials, or human participants.

Before starting the placement, I will confirm with my mentor whether any research-group or computing inductions are required. I will also ensure that relevant Macquarie University WHS and research-integrity requirements are current.

The project will use open-source models, public research code, and researcher-created prompts. Software, model, and dataset licences will be checked before use or redistribution. GPU and remote computing resources will be accessed through authorised accounts.

If required by university policy, I will complete the Assessment of Sanctions Questionnaire (ASQ) before beginning the relevant research activities.

If the project scope later changes to involve human participants, private data, or restricted datasets, the required approvals will be obtained before that work begins.

3 Proposed Timeline

The project is planned around approximately **35 hours of research activity**, consistent with the FOSE7901 half-session workload.

Week	Main activity	Research	Mentor	Deliverable
1	Confirm scope, model, software environment, and selected baseline experiments.	4 h	1 h	1 h
2	Implement the sparse autoencoder method and inspect selected features.	6 h	1 h	2 h
3	Implement JLens and analyse layer-wise concept readouts.	6 h	1 h	2 h
4	Implement activation patching and test interventions across selected layers.	6 h	1 h	2 h
5	Create the controlled prompt set and run baseline model inference.	5 h	1 h	3 h
6	Apply the three methods to selected common prompts and record results.	5 h	1 h	4 h
7	Compare the methods and prepare final figures, report, and presentation.	3 h	1 h	6 h
Total		35 h	7 h	20 h

The project will prioritise successful implementation and comparison over scale. If an experiment requires more computation than expected, the number of prompts will be reduced rather than increasing the overall scope.

Additional models, large datasets, and extensive hyperparameter searches are outside the core scope of Placement 1.

4 Project Data / Records / Creation

The main products of the placement will be the controlled prompt set, model outputs and probabilities, selected internal activations, sparse autoencoder feature results, JLens readouts, activation-patching results, analysis code, notebooks, figures, tables, and an experimental research log.

Data Management Plan

Code and documentation will be managed using Git. Experimental files will be organised into separate locations for prompts, code, notebooks, raw results, processed results, figures, and research records.

For each experiment, I will record the model version, code version, prompt set, method, relevant layer or token position, main parameters, date, and output location. This will make it possible to trace each final result back to the experiment that produced it.

Raw outputs will be kept separate from processed figures and summary tables where practical. Large model weights will not normally be stored in the project repository. Instead, the exact model identifier and source will be recorded.

Successful, unsuccessful, and modified experiments will all be included in the research log. Project

files will be stored using approved university or research computing storage and backed up where appropriate.

No personally identifiable or confidential human data are expected to be produced during Placement 1.

5 Project Deliverable

For the FOSE7901 half-session placement, I propose the following division of the 40% placement deliverable:

Component	Proposed weighting
Research presentation	10%
Research records / reproducibility documentation	5%
Final research report	25%
Total	40%

The research presentation will summarise the motivation, the three implemented methods, and the main preliminary findings.

The research records will include the experimental log, code, model and prompt information, configurations, and key methodological decisions needed to understand how the experiments were conducted.

The final report will describe the implementation of the three baseline methods, their application to the controlled prompt set, the comparison between methods, the main findings and limitations, and recommendations for the next stage of the research.

6 Potential Examiners

Potential examiners will be discussed with my mentor before submission.

Examiner 1: [Name], [School / Department]

Examiner 2: [Name], [School / Department]

References

- Dumas, C., Wendler, C., Veselovsky, V., Monea, G., & West, R. (2025). Separating tongue from thought: Activation patching reveals language-agnostic concept representations in transformers. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 31822–31841.
- Gurnee, W., Sofroniew, N., Pearce, A., Piotrowski, M., Kauvar, I., Chen, R., Soligo, A., Bogdan, P., Ong, E., Wang, R., et al. (2026). Verbalizable representations form a global workspace in language models. *arXiv preprint arXiv:2607.15495*.
- Huben, R., Cunningham, H., Smith, L., Ewart, A., & Sharkey, L. (2024). Sparse autoencoders find highly interpretable features in language models. *International Conference on Learning Representations, 2024*, 7827–7845.